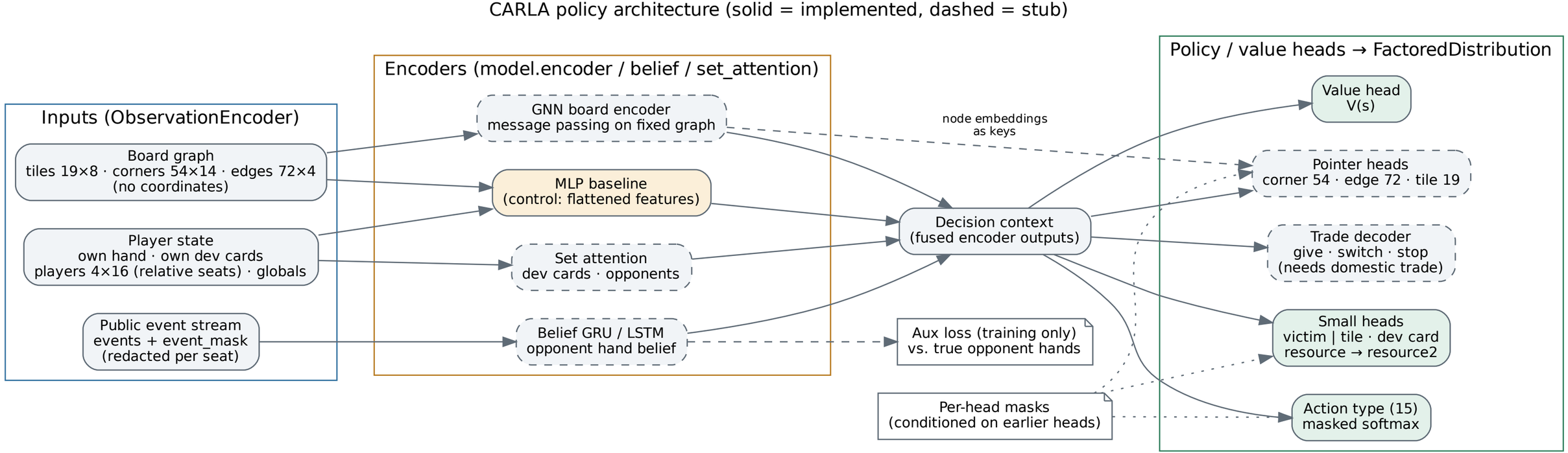
CARLA policy architecture (solid = implemented, dashed = stub)
Inputs (ObservationEncoder)
Board graph
tiles 19×8 · corners 54×14 · edges 72×4
(no coordinates)
Player state
own hand · own dev cards
players 4×16 (relative seats) · globals
Public event stream
events + event_mask
(redacted per seat)
Encoders (model.encoder / belief / set_attention)
GNN board encoder
message passing on fixed graph
MLP baseline
(control: flattened features)
Set attention
dev cards · opponents
Belief GRU / LSTM
opponent hand belief
node embeddings
as keys
Decision context
(fused encoder outputs)
Aux loss (training only)
vs. true opponent hands
Per-head masks
(conditioned on earlier heads)
Policy / value heads → FactoredDistribution
Value head
V(s)
Pointer heads
corner 54 · edge 72 · tile 19
Trade decoder
give · switch · stop
(needs domestic trade)
Small heads
victim | tile · dev card
resource → resource2
Action type (15)
masked softmax